
Education

- 01/2024– **PhD**, *Heidelberg University*, Topic: Statistical Learning in Infinite Dimensions, Expected completion: 12/2027
- 03/2021–10/2023 **M. Sc. Mathematics**, *Heidelberg University*, Final grade: 1.0 (best possible grade)
- 03/2021–06/2023 **M. Sc. Physics**, *Heidelberg University*, Final grade: 1.0 (best possible grade)
- 04/2018–02/2021 **B. Sc. Mathematics**, *KIT*, Karlsruhe, Final grade: 1.1 (best: 1.0)
- 10/2017–02/2021 **B. Sc. Physics**, *KIT*, Karlsruhe, Final grade: 1.1 (best: 1.0)

Research Experience

- 03/2026– **PhD Project**, *Expressivity of Bayesian Neural Networks*, Analyzing approximation and expressivity properties of Bayesian neural networks in function space.
- 01/2025– **PhD Project**, *Bayesian Operator Learning*, Developing a Bayesian operator-learning framework for parameter-dependent PDEs, combining statistical machine learning with numerical computation in Python.
- 01/2024–12/2024 **PhD Project**, *Statistical Learning Theory for Operator Learning*, Theoretical analysis of sample complexity and generalization behaviour for neural operator architectures in infinite-dimensional learning problems.
- 12/2022–10/2023 **Master's Thesis**, *Open-loop stable solutions of a two-legged robot around the standing position*, Formulated and solved multi-phase optimal-control problems and optimized stability for a nonlinear two-legged robot model.
- 02/2022–06/2023 **Master's Thesis**, *Implementation of Modified Gravity Theories with Screening Effects into KFT*, Developed and implemented numerical methods in Python/C++ to evaluate power spectra for modified-gravity models with screening effects.

Publications and Preprints

- 10/2026 **Bayesian Operator Learning: Posterior Existence and Convergence of Point Estimates for Gaussian Priors**, *preprint*.
- 06/2026 **Statistical learning theory for neural operators**, *JMLR*
- 03/2025 **Kinetic field theory: effects of modified gravity theories with screening mechanisms on non-linear cosmic density fluctuations**, *JCAP*.

Teaching and Scientific Activities

- 2024– **Head Assistant**, *Coordinating course organization and supervising more than 150 students and 5 tutors*, “Theory of Deep Learning” (2026, 2024), “Einführung in die Numerik” (2026), and “High-Dimensional Numerics” (2025).
- 2024– **Student Mentoring**, *Supervision of several bachelor's and master's theses*.
- 2024– **Invited Talks and Posters**, *Oberwolfach (2027)*, *SIAM CSE (2027)*, *SciCADE (2026)*, *Swiss Numerics Day (2026)*, *ENUMATH (2025)*, and several workshops.
- 2024– **Peer Reviewer**, *Reviewed papers for ENUMATH proceedings, Bernoulli and Neural Networks*.